

Quiz: Model Fine-Tuning & LLM Optimization

Section 1: Conceptual MCQs

1. What is model fine-tuning?

- A. Training a model from scratch
 - B. Adjusting a pre-trained model on new data
 - C. Only changing prompts
 - D. Increasing model size
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2. Which scenario is best suited for prompting instead of fine-tuning?

- A. Domain-specific chatbot
 - B. Repetitive structured output task
 - C. One-time query with general knowledge
 - D. Training on private dataset
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3. What is the main advantage of fine-tuning over prompting?

- A. Faster inference always
 - B. Better control over task-specific outputs
 - C. No need for data
 - D. Lower memory usage
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4. What does LoRA stand for?

- A. Low Rank Adaptation
 - B. Linear Representation Adjustment
 - C. Layer Optimization and Reuse Algorithm
 - D. Learning Rate Adjustment
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5. In LoRA, what is actually trained?

- A. Entire model weights
 - B. Only input tokens
 - C. Small low-rank matrices
 - D. Output layer only
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6. Why is LoRA considered parameter-efficient?

- A. It reduces dataset size
 - B. It updates only a small subset of parameters
 - C. It removes layers
 - D. It compresses input
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7. What is the role of a tokenizer in fine-tuning?

- A. Train model weights
 - B. Convert text into tokens
 - C. Evaluate model
 - D. Store embeddings
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8. Which step comes first in a fine-tuning workflow?

- A. Model evaluation
 - B. Data collection & cleaning
 - C. Training
 - D. Deployment
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9. What is overfitting in fine-tuning?

- A. Model performs well on all data
 - B. Model memorizes training data but fails on new data
 - C. Model ignores training data
 - D. Model becomes faster
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10. Which metric is commonly used for classification tasks?

- A. BLEU score
- B. Accuracy
- C. Perplexity
- D. Latency

Section 2: True / False

- 11. Fine-tuning always requires large datasets. **False**
 - 12. Prompting does not change model weights. **True**
 - 13. LoRA modifies all layers of the model. **False**
 - 14. Validation data is used to tune hyperparameters. **True**
 - 15. Fine-tuned models can outperform base models on niche tasks. **True**
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Section 3: Scenario-Based Questions

16. Scenario:

You need a chatbot specialized in legal documents with consistent tone and formatting.

👉 What should you choose?

- A. Prompting
 - B. Fine-tuning**
 - C. Zero-shot learning
 - D. Random initialization
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17. Scenario:

You want quick results for generating marketing slogans without training.

👉 Best approach?

- A. Fine-tuning
 - B. Prompt engineering**
 - C. LoRA
 - D. Training from scratch
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18. Scenario:

You have limited GPU resources but want to adapt a large model.

👉 Best method?

- A. Full fine-tuning
- B. LoRA**
- C. Increase batch size
- D. Add more layers

19. Scenario:

Your model performs well on training data but poorly on test data.

👉 What is the issue?

- A. Underfitting
 - B. Overfitting**
 - C. Data leakage
 - D. Tokenization error
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20. Scenario:

You are comparing two models for summarization.

👉 Which metric is most relevant?

- A. Accuracy
 - B. BLEU / ROUGE**
 - C. Latency only
 - D. Token count
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 Section 4: Short Answer

21. Explain one difference between prompting and fine-tuning.

In prompting , uses a pretrained model without changing weights

Fine-tuning: Updates model weights using task-specific data

22. What is one benefit of using LoRA?

Reduces number of trainable parameters → faster and cheaper training

23. List any two steps in a fine-tuning workflow.

1. Tokenization

2. Training the model

24. What is the purpose of validation data?

To evaluate model performance during training and detect overfitting

25. Name one metric used to evaluate LLMs.

Accuracy



Section 5: Practical Thinking

26. You are fine-tuning a model and loss is not decreasing. What could be one reason?

Learning rate is too high or too low

27. Why might fine-tuning be expensive?

Requires large compute (GPU), memory, and training time

28. When should you avoid fine-tuning?

When task can be solved using prompting or small dataset is insufficient

29. What happens if tokenizer is mismatched with the model?

Input tokens become incorrect → model performance degrades or fails

30. How does LoRA reduce computational cost?

Trains low-rank adapter matrices instead of updating full model weights